ECA1403 EMBEDDED SYSTEMS

EXPERIMENT 3

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I2C-BASED HEALTH MONITORING NODE FOR PULSE AND TEMPERATURE ACQUISITION

➔ PROGRAM:

```
#include <Wire.h>

#define LM75_ADDRESS 0x48
#define PULSE_PIN 8

unsigned long pulseCount = 0;
unsigned long lastTempTime = 0;
unsigned long lastBpmTime = 0;

int previousPulseState = LOW;

float readTemperature()
{
  Wire.beginTransmission(LM75_ADDRESS);
  Wire.write(0x00);
  Wire.endTransmission();

  Wire.requestFrom(LM75_ADDRESS, 2);

  if (Wire.available() >= 2)
  {
    byte msb = Wire.read();
    byte lsb = Wire.read();

    int16_t rawTemperature =
```

```
      ((int16_t)msb << 8) | lsb;

    return rawTemperature / 256.0;
  }

  return -999.0;
}

void setup()
{
  Serial.begin(9600);
  Wire.begin();

  pinMode(PULSE_PIN, INPUT);

  Serial.println("I2C HEALTH MONITORING NODE");
  Serial.println("---------------------------");
}

void loop()
{
  unsigned long currentTime = millis();

  /* Pulse counting */

  int currentPulseState = digitalRead(PULSE_PIN);

  if (currentPulseState == HIGH &&
      previousPulseState == LOW)
  {
    pulseCount++;
  }
```

```
previousPulseState = currentPulseState;

/* Temperature acquisition every 1 second */

if (currentTime - lastTempTime >= 1000)
{
  lastTempTime = currentTime;

  float temperature = readTemperature();

  Serial.print("Temperature: ");

  if (temperature > -100)
  {
    Serial.print(temperature, 1);
    Serial.println(" C");
  }
  else
  {
    Serial.println("Sensor Error");
  }
}

/* BPM calculation every 5 seconds */

if (currentTime - lastBpmTime >= 5000)
{
  lastBpmTime = currentTime;

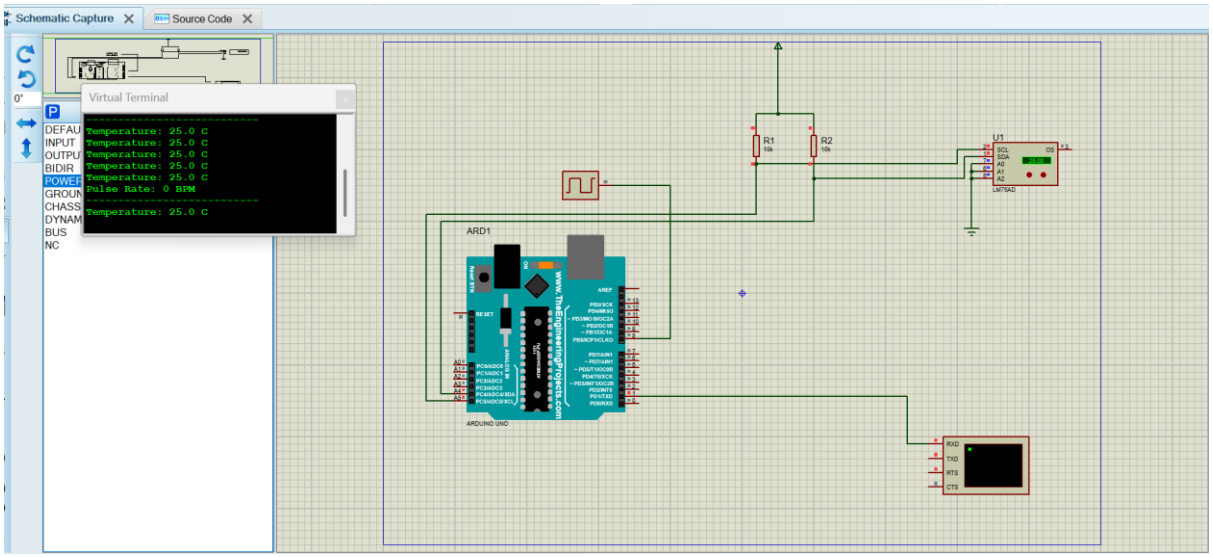
  float bpm = (pulseCount * 60.0) / 5.0;
```

```
    Serial.print("Pulse Rate: ");
    Serial.print(bpm, 0);
    Serial.println(" BPM");

    Serial.println("----------------------------");

    pulseCount = 0;
  }
}
```

OUTPUT:



REFLECTIONS:

Through this experiment, I learned how to design a health monitoring circuit using Arduino Uno in Proteus. I understood the importance of pull-up resistors and power terminals in circuit connections. This experiment improved my knowledge of circuit simulation, I²C communication, and troubleshooting techniques.